

Shubham Gupta

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<https://www.linkedin.com/in/shubhamgupto/> | <https://iamshubhamgupto.github.io>

Software Engineer working on sensor calibration, on-device ML optimization and inference, and audio for wearable and linux embedded devices. Experience in bringing up systems from prototype to production scale.

EXPERIENCE

Rivet Industries, Inc

Software Engineer

Palo Alto, US

Oct 2024 – Present

- Led on-device ML infrastructure and optimizations to significantly reduce accelerator latency.
- Applied QAT algorithms to generate int8 weights using AIMET toolkit for computer vision models.
- Led on-device audio algorithms for spatial filters, noise reduction, wake word, and speech to text LLMs.
- Architected real world to display benchmarking suite to quantify 6DoF tracking and display rendering errors.
- Scaled IMU-magnetometer alignment, soft iron, and IMU thermal calibration algorithms from prototype to EVT builds.

M²LInES, New York University

Research Associate | Advisor: *Dr. Laure Zanna*

New York City, US

Aug 2024 – Oct 2024

- Eliminated inter-epoch synchronization bottlenecks using Nsight Systems, saving 2 hours/day on NERSC & NYU clusters.
- Optimized GPU residency via asynchronous data prefetching and worker persistence, reducing I/O stalls.
- Implemented Mixed Precision Training, reducing GPU usage by 64% during the training phase while maintaining skill.

NYU ARC Robotics

Computer Vision and DevOps Lead | Mentor: *Chris DiMauro*

New York City, US

Sep 2022 – Jun 2024

- Secured 5th place in RMNA 2023 internationally against elite robotics teams from USA, Japan and Singapore.
- Led large scale HPC experimentation for training and PTQ tuning int16 quantized object detection models.
- Introduced containerization and led the OTA deployment of controls, sensors, and model inference microservices.
- Established two-way data connections using UART, Ethernet protocol between devices - Nvidia Jetson and Arduino.

HPE Aruba Networking

Software Development Intern

Bangalore, India

Jan 2022 – Jul 2022

- Developed SpringBoot APIs integrating Elasticsearch and Postgres data for edge-to-cloud IoT Operations.
- Designed and deployed a monitoring dashboard for 150k customer device statistics using Elasticsearch, Grafana, and SQL.

Indian Space Research Organization

Research Intern | Advisor: *Dr. Hebbar R*

Bangalore, India

Aug 2019 – Apr 2020

- Managed data annotations and experiments, published findings as lead author at the CVMI 2022 conference.
- Built an automated Sentinel-2 data pipeline with DeepLabv3+ and SnaPy/GDAL for water body segmentations.

EDUCATION

New York University

Master of Science in Computer Engineering — 3.8/4.00

New York City, US

PES University

Bachelor of Technology in Computer Science and Engineering — 8.39/10.0

Bangalore, India

PUBLICATIONS

Dheeshjith, S., Subel, A., **Gupta, S.**, Adcroft, A., Fernandez-Granda, C., Busecke, J., & Zanna, L. (2024). Transfer Learning for Emulating Ocean Climate Variability across CO_2 forcing. arXiv preprint arXiv:2405.18585. [\[preprint\]](#) [\[ICML Oral 2024\]](#)

Lee, K., **Gupta, S.**, Kim, S., Makwana, B., Chen, C., & Feng, C. (2023). SO-NeRF: Active View Planning for NeRF using Surrogate Objectives. arXiv preprint arXiv:2312.03266. [\[preprint\]](#) [\[code\]](#)

Gupta, S., Uma, D., & Hebbar, R. (2023). Analysis and application of multispectral data for water segmentation using machine learning. In Computer Vision and Machine Intelligence: Proceedings of CVMI 2022 (pp. 709-718). Singapore: Springer Nature Singapore. [\[preprint\]](#) [\[code\]](#) [\[publication\]](#) [\[slides\]](#)

PROJECTS

Cavemen: A prehistoric approach for Mapless Navigation | [\[Website\]](#) | *Python3, SIFT, FAISS, VBoW, Redis* 2023

- Visual Place Recognition and navigation using only monocular input and classic visual features. **Top submission** for coursework Robot Perception ROB-GY 6203 Fall 2023. [\[LinkedIn\]](#)

LYCB: Leave Your Clothes Behind | [\[Website\]](#) | *HPC, Segment-Anything, COLMAP, Pytorch3D* 2023

- Proof-of-Concept to capture real world clothes on hand held devices and create assets to be used in Blender or virtual try-ons.

TECHNICAL SKILLS

Languages: Python3, C++, Java, L^AT_EX

Libraries: ONNX, TensorRT, QAIRT, FastAPI, ADB, Hydra, Slurm

Applications: NVIDIA Nsight, Docker, Elasticsearch, SQL, Grafana